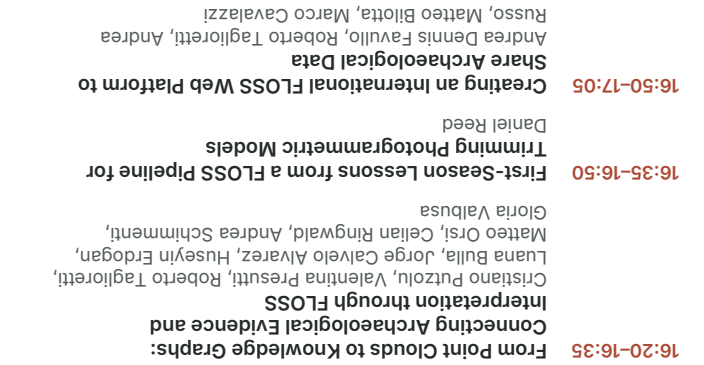
ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA | DEPARTMENT OF HISTORY AND CULTURES

20th EDITION

ARCHEO.FOSS 2026

International conference on open software, hardware, processes, data and formats in archaeological research



30 November – 1 December 2026

VENUE

Department of History and Cultures (DiSCi)
Alma Mater Studiorum – Università di Bologna
Piazza San Giovanni in Monte 2, 40124 Bologna, Italy

Conference programme

www.archeofoss.org

Practical information

Venue
Dipartimento di Storia Culture Civiltà, Alma Mater Studiorum Università di Bologna, Piazza San Giovanni in Monte 2, 40124 Bologna, Italy

Rooms
Aula Prodi, Aula Specola, Aula Morandi, Aula Capitani, Aula Gambi

Hosted by
DArchLab – Digital Archaeology Laboratory, Dipartimento di Storia Culture Civiltà, Alma Mater Studiorum Università di Bologna

With the contribution of
Dipartimento di Storia Culture Civiltà, Alma Mater Studiorum Università di Bologna

Contact
darchlab@unibo.it

Full programme online

www.archeofoss.org/2026/programme

At a glance

30 November 2026

09:00–09:15	Aula Prodi · Welcome addresses
09:15–09:30	Aula Prodi · Opening remarks
09:30–11:15	Aula Prodi · Panel: OsteoArchaeology FLOS
09:30–12:10	Aula Specola · Panel: FLOSS Geospatial Computing in Archaeology: Spatial Analysis, Modelling and Reproducible Workflows
11:15–13:30	Aula Prodi · Panel: Artificial Intelligence in Archaeological Research: State of the Art, Limitations, and Advanced Applications
12:10–14:55	Aula Specola · Panel: Protocols for remote sensing in cultural heritage diagnostics
13:30–15:30	Aula Prodi · Panel: From Fragmented Data to Interoperable Knowledge: Semantic Technologies, Ontologies and Linked Open Data in Archaeology
13:30–14:45	Aula Capitani · Panel: Artificial Intelligence and Digital Technologies for the Communication of Archaeological Heritage
15:15–16:50	Aula Morandi · Panel: Archaeological Landscapes as Socio-Ecological Systems: FLOSS-Based Modelling and Interpretation
15:30–17:30	Aula Prodi · Panel: Unlocking the Canopy, Locking the Data? LiDAR Archaeology and Its Quest for Open Science
17:30–18:00	Aula Prodi · Book presentation
18:00–19:00	Aula Prodi · Keynote lecture

1 December 2026

09:00–10:45	Aula Prodi · Panel: Beyond Visualisation: Assessing Uncertainty and Verifying Geometry in Hypothetical Virtual 3D Reconstructions
10:45–11:30	Aula Prodi · Keynote lecture
11:30–12:15	Aula Prodi · Round-table discussion
12:15–14:15	Aula Prodi · Panel: FLOSS Geospatial Computing in Archaeology: Spatial Analysis, Modelling and Reproducible Workflows
13:30–15:00	Aula Gambi · Workshop
14:15–16:15	Aula Prodi · Panel: FOSS beyond research: open-source solutions in production for professional archaeology and heritage management
15:00–17:00	Aula Morandi · Workshop
16:15–18:45	Aula Prodi · Panel: From Fieldwork to Dissemination through FLOSS and FLOSH

1 December 2026

Aula Prodi

Panel: Beyond Visualisation: Assessing Uncertainty and Verifying Geometry in Hypothetical Virtual 3D Reconstructions
Chair and introduction by Fabrizio Ivan Apollonio, Federico Fallavollita, Riccardo Foschi, Krzysztof Koszewski, Emanuel Demetrescu, Sander Münster
09:00–09:05 Panel presentation

09:05–09:20 An attempt at a hypothetical 3D reconstruction of one of the 13 Djeddar Funerary Mausoleums in Tiaret, Algeria Using Aerial Photogrammetry.
Kamel Baker

09:20–09:35 Scientific Reliability of Virtual Reconstructions: Uncertainty Assessment and Geometric Validation of Algeria's Archaeological Heritage — The Cases of Tlmgad and Djemila
Khaoula Bertal

09:35–09:50 The archaeologist at the core of the Digital Workflow: AI, 3D architecture, and epistemological transparency for the reconstruction of Dur-Kurigalzu (Baghdad, Iraq)
Antonio Lanzetta

09:50–10:05 Beyond Visualisation: Assessing Uncertainty and Verifying Geometry in Hypothetical Virtual 3D Reconstructions
Fabrizio Ivan Apollonio, Federico Fallavollita, Riccardo Foschi, Krzysztof Koszewski, Emanuel Demetrescu, Sander Münster

10:05–10:30 Discussion

10:30–10:45 Coffee break

Keynote lecture

10:45–11:30 Aprire la conoscenza archeologica: continuità e trasformazioni dell'editoria scientifica. Lo sguardo di Archeologia e Calcolatori
Alessandra Caravale (ISPCR - CNR)

Round-table discussion

11:30–12:15 Scavi al microfono, tra divulgazione e creazione di comunità
Chiara Boracchi (Università degli Studi di Milano "La Statale")
Discussants: Marina Lo Blundo (Parco archeologico di Ostia antica), Sebastian Paolo Righi (MasterPast Scientific Committee), Giuliano Volpe (Università degli Studi di Bari "Aldo Moro")

12:15–12:30 Moderato
Chair and introduction by Carlo Bicchieral, Jacopo Celani, Marco Thya W.B. van den Berg, Jane Bunting
Panel: FLOSS Geospatial Computing in Archaeology: Spatial Analysis, Modelling and Reproducible Workflows

12:15–12:30 Putting lives in context: geospatial reconstructions of landcover from pollen records using the Multiple Scenario Approach
Taylor Zaneri

12:30–12:45 Lunch break

12:30–12:45 Open Source Landscape Analysis and Suitability Modelling in GIS
Carlo Bicchieral

13:30–13:45 Between plains and mountains. Comparison between results on application of Maximum Entropy algorithm on Bassa Romandiola and Monti della Laga Dataset
Cristiano Putzolu

13:45–14:00 Chair and introduction by Marco Cavalazzi, Tyler Duane Johnson, Francesco Alfinito
An integrated map of microrelief and deposit thickness: the case of Cervia Vecchia (RA)

14:00–14:15 Discussion

Panel: FOSS beyond research: open-source solutions in production for professional archaeology and heritage management

14:15–14:20 Chair and introduction by Andrea Simeoni
Panel presentation

14:20–14:35 QGIS as a FLOSS Workbench for Multisensor Analysis in Preventive Archaeology: Hybrid Workflows for Infrastructure Design
Fabio Nieddu, Pina Maria Derudas, Pierfrancesco Izzo

14:35–14:50 QGIS e analisi archeologiche avanzate. Il caso di FS Engineering
Francesca Frandi, Davide Pagliarosi, Giancarlo Pastura, Matteo Serpetti

14:50–15:05 FLOS for Hegra Conservation Project: QGIS, PostGIS and collaborative offline workflows
Nicoletta Capanna, Maurizio Dragoni

15:05–15:20 MAGOH Spell: un plugin QGIS FLOSS per l'interoperabilità dei dati archeologici territoriali
Francesca Anichini, Lorena Bravi, Francesco D'Antoni, Gabriele Gattiglia

15:20–15:35 Archaeo.pro: an open, offline-first toolchain for the sorveglianza archeologica, with the archive owned by the archaeologist
Marco Montanari

15:35–16:00 Discussion

16:00–16:15 Coffee break

Panel: From Fieldwork to Dissemination through FLOSS and FLOSH
Chair and introduction by Marco Cavalazzi, Tyler Duane Johnson, Cristiano Putzolu
16:15–16:20 Panel presentation

16:20–16:35 From Point Clouds to Knowledge Graphs: Connecting Archaeological Evidence and Interpretation through FLOSS
Cristiano Putzolu, Valentina Presutti, Roberto Tagliaretti, Luana Bulla, Jorge Calvelo Alvarez, Huseyin Erdogan, Matteo Orsi, Celian Ringwald, Andrea Schimmenti, Gloria Valbusa

16:35–16:50 First-Season Lessons from a FLOSS Pipeline for Trimming Photogrammetric Models
Daniel Reed

16:50–17:05 Creating an International FLOSS Web Platform to Share Archaeological Data
Andrea Dennis Favullo, Roberto Taglioretti, Andrea Russo, Matteo Biotta, Marco Cavalazzi

17:05–17:20 Publishing an excavation archive without a server: GeoContext and GeoContext Storybook
Marco Montanari, Silvia Bernardoni

17:20–17:35 SITAR Portal Geoservices: Strategies for Sharing Strategic Data for Land Planning and Landscape Protection between Open Data and FLOSS Approaches
Mirella Serlorenzi, Rocco Bochicchio, Ascanio D'Andrea, Maura Fadda, Paolo Rosati, Daniele Sepio, Alessandro Vecchione

17:35–17:50 Applicazioni open source a modelli digitali in siti archeologici romani della Campania: Piscina Mirabilis; Villa Arianna.
Alfredo Trocciola, Raffaele Pica, Alessandra Esposito, Karen Holmberg, Renato Somma

17:50–18:15 Discussion

18:15–18:45 Closing remarks

Aula Gambi

Workshop
13:30–15:00 Making Uncertainty Visible: Methods and Tools for Hypothetical 3D Heritage Reconstructions
Riccardo Foschi

To register for the workshop, please send an email to cowher.workshop@gmail.com, indicating the workshop title and your full name.

Aula Morandi

Workshop
15:00–17:00 Documentation and publication of 3D outcomes: Methods and Tools for Hypothetical 3D Heritage Reconstructions
Igor Piotr Bajena

To register for the workshop, please send an email to cowher.workshop@gmail.com, indicating the workshop title and your full name.

ARCHEO.FOSS 2026

Conference programme · 30 November – 1 December 2026 · Bologna

30 November 2026

Aula Prodi

Welcome addresses

09:00–09:15 **Roberto Balzani (Director of DiSCi), Nicolò Marchetti (Archaeology Section Coordinator)**

Opening remarks

09:15–09:30 **Gabriele Gattiglia (President of ArcheoFOSS)**

Panel: OsteoArchaeology FLOS

Chair and introduction by Alessandra Mazzucchi, Roberto Taglioretti

09:30–09:35 Panel presentation

09:35–09:50 **Optimising Accessible 3D Pipelines for In Situ Osteological Recovery: Methodological Bottlenecks and FLOS Integration**
Matteo Orsi, Gloria Valbusa, Roberto Taglioretti, Maurizio Marinato, Alessandra Mazzucchi

09:50–10:05 **Statistical software jamovi applied to bioanthropological data**
Matilde Vianello, Maurizio Marinato, Andrea Russo, Alberto Zanatta, Alessandra Mazzucchi

10:05–10:20 **Filling gaps: a FLOS atlas on non-adult bones with MAPOD4D®**
Maria Elena Tarantino, Davide Carraro, Caterina Giostra, Maria Fortunati, Roberto Taglioretti, Alessandra Mazzucchi

10:20–10:35 **La morte ti fa colt@**
Susanna Saderi, Ryhor Pauliuts, Maria Rosaria Luberto

10:35–10:50 Coffee break

10:50–11:15 Discussion

Panel: Artificial Intelligence in Archaeological Research: State of the Art, Limitations, and Advanced Applications

Chair and introduction by Andrea Fiorini, Maurizio Cattani, Giuseppe Guarino, Francesco Iacono, Panos Kaplanis, Doukas Vomvyras

11:15–11:20 Panel presentation

11:20–11:35 **Is AI reorienting archaeological methods?**
Roberto Ragno, Lorenzo Cardarelli

11:35–11:50 **Accountable, not fluent: harris-mcp, an open Model Context Protocol server for LLM-assisted reading of Harris matrices**
Marco Montanari

11:50–12:05 **Beyond democratisation claims: AI, expertise and productive friction in AUTOMATA**
Nevio Dubbini, Gabriele Gattiglia

12:05–12:20 **Text in Space, Truth in the Schema: A FOSS Neuro-Symbolic Middleware for Querying Roman Archaeology without Hallucinations (IN-ROME Project)**
Eleonora Iacopini

12:20–12:45 Discussion

12:45–13:30 Lunch break

Panel: From Fragmented Data to Interoperable Knowledge: Semantic Technologies, Ontologies and Linked Open Data in Archaeology

Chair and introduction by Fabrizio Ducati, Celian Emile Roger Ringwald, Huseyin Erdo an

13:30–13:35 Panel presentation

13:35–13:50 **Building and disseminating a controlled and free vocabulary for archaeology, between open access policy, compliance with standards and multilingualism: the example of the Pactols thesaurus**
Blandine Nouvel

13:50–14:05 **From Fragmented Objects to Federated Knowledge: FAIR Digital Objects as FDOx for 3D Models and Software in the Archaeological LOD Cloud**
Allard Mees, Florian Thiery

14:05–14:20 **R: The repository is the database: OntoRAG, ontology-governed retrieval over archaeological narrative sources**
Marco Montanari

14:20–14:35 **X-DigCheck: Co-Evolving Application Profiles and Knowledge Graphs, Demonstrated on the RTI Documentation of Rupe Magna**
Celian Ringwald, Huseyin Erdogan, Valentina Presutti, Cristiano Putzolu

14:35–14:50 **From Digital Heritage to Connected Data: Automating Geographical Metadata and Bibliographic Knowledge Graphs in Archeologia e Calcolatori**
Alessandra Caravale, Antonio D'Eredità, Giacomo Mancuso, Paola Moscati, Nicolò Paraciani

14:50–15:05 **More Than Just a Pretty Vase: How Kerameikos.org Makes Studying Greek Pottery Accessible, Connected, and Collaborative through Linked Open Data**
Tyler Jo Smith, Ethan Gruber, Abigail Bradford

15:05–15:30 Discussion

Panel: Unlocking the Canopy, Locking the Data? LiDAR Archaeology and Its Quest for Open Science

Chair and introduction by Michele Abballe, Paolo Forlin, Giacomo Vinci

15:30–15:35 Panel presentation

15:35–15:50 **Intricate palimpsests. Disentangling Italian cultural landscapes through open and drone-borne LiDAR**
Giacomo Vinci

15:50–16:05 Coffee break

16:05–16:20 **From Flight Planning to DEM Visualisation: Improving UAV LiDAR Data for archaeological survey in Mountain Environments**
Jorge Hernan Gonzalez Perez

16:20–16:35 **Mapping the Alpine Past: LiDAR Data Accessibility and Open-Source Approaches in the Belluno Dolomites**
Paolo Forlin

16:35–16:50 **Open methods, governed coordinates: a FLOSS workflow for responsible LiDAR sharing in the Lower Amazon**
Ney Gomes

16:50–17:05 **Automated Reading of Flattened Alluvial Landscapes: A Comparative Application of Open LiDAR DTMs in Emilia-Romagna**
Michele Abballe

17:05–17:30 Discussion

Book presentation

17:30–18:00 **Archeologia e Calcolatori, Supplemento 14, Surveying in archaeology in the digital era: Rethinking old procedures and imagining new standards · Edited by Marco Cavalazzi, Tyler Duane Johnson and Cristiano Putzolu**
Alessandra Caravale (ISPCR-CNR), Gabriele Gattiglia (Università di Pisa), Valentina Presutti (Alma Mater Studiorum Università di Bologna)

Keynote lecture

18:00–18:45 **Critical Software Practice in Archaeology**
Jeremy Huggett (University of Glasgow)

18:45–19:00 Day-one closing remarks

Aula Specola

Panel: FLOSS Geospatial Computing in Archaeology: Spatial Analysis, Modelling and Reproducible Workflows

Chair and introduction by Carlo Bicchierai, Jacopo Celani, Marco Moderato

09:30–09:35 Panel presentation

09:35–09:50 **From Legacy Data to Spatial Models: A Reproducible FLOSS Workflow for Early Medieval Ceramic Imports in Northern Italy (6th–8th Centuries AD)**
Jacopo Celani

09:50–10:05 **Unpacking the type: attribute-level network modelling of the Lunigiana statue-stelae in R**
Dario Nincheri, Luca Biancalani

10:05–10:20 **La modellazione degli spazi liturgici calpestabili. Un nuovo plugin QGIS per l'integrazione delle fonti scritte con i dati archeologici, con il fine di ricostruire la mobilità relativa al triportico della prima Basilica cristiana del Santo Sepolcro.**
Eleonora Passannanti

10:20–10:35 **Testing Settlement–Mobility Relationships with "From Everywhere To Everywhere" Least-Cost Paths and Monte Carlo Simulations: a case study from Hellenistic Armenia**
Federico Izzi

10:35–10:50 Coffee break

10:50–11:05 **Landscape Legibility within a funerary landscape: a GIS-based quantitative analysis of tomb visibility and human movement in Wadi Halfayin, Ad-Dakhiliyah, Sultanate of Oman**
Theodore Sheehan

11:05–11:20 **Geospatial Analysis of Storage Structures of the Mesolithic-Neolithic Transition in Britain and Ireland**
Raúl Bagüés-Brocal, Valentina Tumolo, Marco Moderato, Marco Madella

11:20–11:35 **Hillforts and Pastoral Landscapes in the Fucino Basin: A GIS Approach to Transhumance in the Central Apennines**
Elena Scarsella

11:35–11:50 **Landscape, mobility and seasonality of the Southern Neolithic agro-pastoralists in India**
Milica Rajicic, Marco Moderato, Marco Madella

11:50–12:10 Discussion

Panel: Protocols for remote sensing in cultural heritage diagnostics

Chair and introduction by Gloria Valbusa, Matteo Orsi

12:10–12:15 Panel presentation

12:15–12:30 **Temperature in scanning: could thermal radiation interfere with acquisition?**
Gloria Valbusa, Roberto Taglioretti, Matteo Orsi

12:30–12:45 **Multi-Scale Trajectory Optimisation for Ground-Based SLAM in Archaeological Contexts: Mitigating Noise and Maximising Data Yield Across Confined and Open Excavations**
Matteo Orsi, Gloria Valbusa, Giovanni Castelli, Luca Carabelli, Alessandra Mazzucchi, Roberto Taglioretti

12:45–13:30 Lunch break

13:30–13:45 **An Open-Source Workflow for Recording the Excavation Process: Volumetric Reconstruction from iPad LiDAR Data at Ca' Nova di Albareto (Parma)**
Jorge Hernan Gonzalez Perez, Erik Zanotti, Cristiano Putzolu

13:45–14:00 **Beyond Default Topologies: Optimising the Meshroom Pipeline for High-Fidelity 3D Point Cloud Reconstruction and Acquisition Protocol Redesign**
Matteo Orsi, Gloria Valbusa, Andrea Russo, Alessandra Mazzucchi, Roberto Taglioretti

14:00–14:15 **Decoding the Palimpsest: An Open Computational Imaging Protocol for Weathered Petroglyphs at Shokur, Oman**
Kristina Zurzic, Nunzia Larosa, Francesco Genchi

14:15–14:30 **Multi-sensor Diagnostic Protocol in areas up to 3×2 m: FLOSS Workflows, Repeatability, and the Faro Convention**
Matteo Orsi, Gloria Valbusa, Filippo di Marco, Roberto Taglioretti

14:30–14:55 Discussion

Aula Morandi

Panel: Archaeological Landscapes as Socio-Ecological Systems: FLOSS-Based Modelling and Interpretation

Chair and introduction by Francesca Cornella, Filippo Brandolini, Francesco Carrer, Enrico Croce

15:15–15:20 Panel presentation

15:20–15:35 **From geosystem to territory: theoretical framework of a GIS-based model of agricultural potential in the Neolithic Pyrenees**
Guillem Salvador-Baiges, Ferran Antolin, Ermengol Gassiot-Ballbè

15:35–15:50 **Agriculture and subsistence in mountain sites: a FLOSS approach to the reconstruction of the subsistence economy of the Bronze Age settlement of Monte della Croce (Bologna, Italy)**
Diletta Panno

15:50–16:05 Coffee break

16:05–16:20 **Agricultural strategies, settlement sustainability and water dynamics in the ancient Indus Valley**
Blanca Rodríguez Magem, Marco Moderato, Marco Madella

16:20–16:35 **Territory and Subsistence: A Preliminary Archaeobotanical and Landscape Approach to Ca' Nova di Albareto between the Bronze Age and Iron Age**
Roberta Matranga

16:35–16:50 Discussion

Aula Capitani

Panel: Artificial Intelligence and Digital Technologies for the Communication of Archaeological Heritage

Chair and introduction by Melania Marano

13:30–13:35 Panel presentation

13:35–13:50 **Adaptive Communication in Cultural Heritage Conversational Agents: A Prompt-Based Approach for User Expertise-Aware Responses**
Gioele Litrico, Giovanni Gallo, Filippo Stanco

13:50–14:05 **Giving a Voice to the Altamura Man: Artificial Intelligence in Museum Engagement**
Simone Favaron

14:05–14:20 **From excavation data to public narrative: Artificial Intelligence as a bridge in the communication of archaeological stratigraphy**
Melania Marano

14:20–14:45 Discussion

The programme of 1 December 2026 and the workshops are on the three panels of the other side.